Jordan Watkins, PhD

Mathematics, physics & CS instruction — PhD, published researcher

(708) 256-5771 · incohatus.none@gmail.com · github.com/incnone · LinkedIn · Charlottesville, VA · relocating

PhD mathematician with four years of university teaching, a published research record, and tutoring experience across mathematics, physics, and computer science. Taught calculus and assisted differential equations at the University of Michigan; comfortable from single- and multivariable calculus through topology, abstract algebra, and statistics, and in C++ and Python for computer-science students. Active in a research seminar and seeking lecturer, instructor, adjunct, or tutoring work.

TEACHING & TUTORING EXPERIENCE

Graduate Student Instructor, Mathematics · University of Michigan

2007–2011

- Independently taught multiple sections of Calculus I and Calculus II — lecturing, assessment, and grading
- Teaching assistant for Differential Equations
- Led required weekly tutoring sessions in single- and multivariable calculus, differential equations, and basic probability
- Mentored individual paying students, including in linear algebra

Mathematics & Physics Tutor · Independent

Intermittently since 2005

- Tutored across a wide range: topology, abstract algebra, differential equations, statistics, linear algebra, and electricity & magnetism
- Supported computer-science students in C++
- Graduate-school teaching included weekly tutoring in calculus, differential equations, and probability, plus independent paying students

SUBJECTS

Calculus I–III · Differential equations · Linear algebra · Abstract algebra · Topology · Probability & statistics · Electricity & magnetism · C++ · Python

EDUCATION

Ph.D. in Mathematics · University of Michigan

2013

Specialization: differential geometry (dynamics on Riemannian manifolds). Thesis: “The Rank Rigidity Theorem for Manifolds With No Focal Points.”

B.A. in Mathematics and Physics · Knox College

2007

Valedictorian

PUBLICATIONS

“The Higher Rank Rigidity Theorem for Manifolds With No Focal Points” · *Geometriae Dedicata*, vol. 164

Aug 2012 · pp. 319–349

Springer · arXiv:1111.5428

CONTINUED ENGAGEMENT

Mathematical Physics Seminar · University of Virginia

2024–Present

Regular participant. Ongoing engagement with current research in geometry and mathematical physics.

Erdős Institute — Data Science Boot Camp · coursework

Applied data science in Python (pandas, scikit-learn).

TECHNICAL BACKGROUND FOR CS TEACHING

Co-Developer — *Bean and Nothingness* · RBOR Games

2020–2021 · shipped Dec 2021

store.steampowered.com/app/1706090/Bean_and_Nothingness/

Logic-puzzle game · 98% positive from 198 Steam reviews · prototype work from 2012

- Co-lead programmer (C++, SDL, OpenGL): replay/undo state system, GUI level editor shipped via Steam Workshop, custom sound mixer, core game-loop and entity logic
- Lead puzzle designer: created 100+ and edited all ~200 puzzles

Competitive Rating & Matchmaking Systems · CoNDOR · open source

multiple seasons, through 2022

github.com/incnone/condor_ratings

Bayesian rating model, outcome forecasting, and ILP matchmaking for a multi-league season (Python, C++, MySQL, PuLP)

- Custom Bayesian paired-comparison model (Gaussian priors + binomial evidence) fit by MAP estimation with hand-derived gradients and a hand-written optimizer; prior variance chosen by held-out log-likelihood; heavy-tailed link function with its tail weight fit to past results
- Best-of-3 / best-of-5 outcome forecasts from jointly normal ratings with covariance — exposed by patching a C++ Bayesian Elo engine
- Weekly matchmaking as an integer linear program maximizing predicted-outcome entropy under degree constraints

Architect & Maintainer — Necrobot · Open source

2015–Present

github.com/incnone/necrobot

Discord bots managing competitive races and league seasons; in continuous community use for 10+ years

- One layered codebase (~150 modules, 16 packages, documented acyclic dependencies) running two production bots
- Async Python with MySQL persistence, Google Sheets integration, and TrueSkill-based (Bayesian) ratings

SKILLS

Teaching	Calculus, differential equations, linear algebra, probability, physics; section instruction and one-on-one
Programming for courses	C++, Python, SQL
Research	Differential geometry; rigidity on Riemannian manifolds
Tools	Git, MySQL, Visual Studio

Career note: independent study and freelance tutoring 2013–2020, during part of which I carried family caregiving responsibilities.